16

Baking and roasting

Baking and roasting are essentially the same unit operation: they both use heated air to alter the eating quality of foods. The terminology differs in common usage; baking is usually applied to flour-based foods or fruits, and roasting to meats, nuts and vegetables. In this chapter the term *baking* is used to include both operations. A secondary purpose of baking is preservation by destruction of micro-organisms and reduction of the water activity at the surface of the food. However, the shelf life of most baked foods is short unless it is extended by refrigeration or packaging.

16.1 Theory

Baking involves simultaneous heat and mass transfer; heat is transferred into the food from hot surfaces and air in the oven and moisture is transferred from the food to air that surrounds it and then removed from the oven (Chapter 1, Section 1.2 and Fig. 1.4).

In an oven, heat is supplied to the surface of the food by a combination of infrared radiation from the oven walls, by convection from circulating air and by conduction through the pan or tray on which the food is placed. Infrared radiation (Chapter 18) is absorbed into the food and converted to heat. Air, other gases and moisture vapour in the oven transfer heat by convection. The heat is converted to conductive heat at the surface of the food. A boundary film of air acts as a resistance to heat transfer into the food and to movement of water vapour from the food. The thickness of the boundary layer is determined mostly by the velocity of the air and the surface properties of the food (Chapters 1 and 15) and in part controls the rates of heat and mass transfer. Convection currents promote uniform heat distribution throughout the oven, and many commercial designs are fitted with fans to supplement natural convection currents and to reduce the thickness of boundary films. This increases heat transfer coefficients and improves the efficiency of energy utilisation.

Heat passes through the food by conduction in most cases, although convection currents are established during the initial heating of cake batters (Mizukoshi, 1990). The low thermal conductivity of foods (Chapter 1, Table 1.5) causes low rates of conductive

heat transfer and is an important influence on baking time. Conduction of heat through baking pans or trays increases the temperature difference at the base of the food and increases the rate of baking compared to the surface crust. The size of the pieces of food is an important factor in baking time as it determines the distance that heat must travel to bake the centre of the food adequately. Methods of heat transfer and resistances to heat and mass transfer are discussed further in Chapter 1. The technology of breadmaking is described in detail by Cauvain and Young (1998) and the technology of cake making by Bennion and Bamford (1997).

When a food is placed in a hot oven, the low humidity of air in the oven creates a moisture vapour pressure gradient, which causes moisture at the surface of the food to evaporate and this in turn creates movement of moisture from the interior of the food to the surface. The extent of moisture loss is determined by the nature of the food, movement of air in the oven and the rate of heat transfer. When the rate of moisture loss from the surface exceeds the rate of movement from the interior, the zone of evaporation moves inside the food, the surface dries out, its temperature rises to the temperature of the hot air (110–240°C) and a crust is formed. Because baking takes place at atmospheric pressure and moisture escapes freely from the food, the internal temperature of the food does not exceed 100°C. These changes are similar to those in hot-air drying (Chapter 15), but the more rapid heating and higher temperatures used in baking cause complex changes to the components of the food at the surface (Section 16.3). These changes both enhance eating qualities and retain moisture in the bulk of the food. In contrast with dehydration, where the aim is to remove as much water as possible with minimal changes in sensory quality, in baking the heat-induced changes at the surface of the food and retention of moisture in the interior of some products (cake, bread, meats, etc.) are desirable quality characteristics. In other products, such as biscuits and crispbread, loss of moisture from the interior is required to produce the desired crisp texture.

The types of mass and heat transfer in different parts of a food during baking are described in Table 16.1.

Equations for the calculation of heat transfer during baking are described in Chapter 1 and a relevant sample problem is given in Chapter 18.

Energy consumption during baking is of the order of 450–650 kJ per kilogram of food. Most of the heat is used to heat the food, to evaporate water, to form the crust, to superheat water vapour (steam) that is transported through the crust and to heat the dry crust. Commercial ovens are insulated with up to 30 cm of mineral wool, refractory tiles or similar materials, and heat losses are therefore minimised. Other energy conservation devices are described in Section 16.2 and Chapter 1 (Section 1.4.4).

Table 16.1 Mass and heat transfer during baking

Zone in the food	Type of mass transfer	Type of heat transfer
Boundary layer	Vapour diffusion	Conduction, convection, radiation
Crust	Vapour diffusion	Conduction, vapour movement (convection)
Evaporation zone	Vapour diffusion, surface diffusion, capillary flow	Conduction, movement of vapour and liquid water
Interior	Capillary flow	Conduction

Adapted from Hallstrom and Skjoldebrand (1983).

16.2 Equipment

Detailed descriptions of the different stages in baking are described by Manley (1998). Equipment used to prepare foods prior to baking is described in Chapter 5 (mixing and forming). In this chapter, fuel fired and electric ovens are described. Microwave and dielectric ovens are described in Chapter 18. Nearly all of the oven designs below now incorporate advanced energy saving features and microprocessor controls (Chapter 2). Pre-programmed baking controls enable operators to select a product code without the need to remember baking settings. Microprocessor control of the sequence, duration, temperature and humidity of baking prevents operator error and the use of incorrect baking conditions. Additionally, ovens at different sites can be programmed identically to achieve product uniformity from all factories. Changes to control settings or introduction of new products can be made using a diskette or from a central office by either a direct modem connection to the oven or via a communications network using a central computer.

Ovens are classified into direct or indirect heating types.

16.2.1 Direct heating ovens

In directly heated ovens, air and the products of combustion are recirculated by natural convection or by fans. The temperature in the oven is controlled automatically, by adjustment of air and fuel flow rates to the burners. Natural gas is commonly used, but propane, butane, fuel oil or solid fuels are also found (Chapter 1). Gas is burned in ribbon burners located above and below conveyor belts in continuous ovens, and at the base of the cabinet in batch ovens. Safety features are incorporated to extinguish the burners automatically if abnormal baking conditions arise, and pressure-relief panels are fitted to the top of the ovens to protect personnel should a gas explosion occur.

The advantages of direct heating ovens include:

- short baking times
- high thermal efficiencies
- good control over baking conditions (using the fan speed and the rate of fuel consumption)
- rapid start-up, as it is only necessary to heat the air in the oven.

However, care is necessary to prevent contamination of the food by undesirable products of combustion, and gas burners require regular servicing to maintain combustion efficiency. Microwave and dielectric ovens (Chapter 18) are another example of direct-heating ovens.

16.2.2 Indirect heating ovens

Steam tubes are either heated directly by burning fuel or supplied with steam from a remote boiler (Fig. 16.1). The steam tubes then heat air in the baking chamber. Heated air is commonly recirculated through the baking chamber and through a separate heat exchanger (Fig. 16.2). Alternatively, combustion gases are passed through banks of radiator tubes in the baking chamber, or fuel is burned between a double wall and the combustion products are exhausted from the top of the oven. Electric ovens are heated by induction heating radiator plates or bars. In batch ovens, the walls and base are heated whereas in continuous ovens, heaters are located above, alongside and below a conveyor

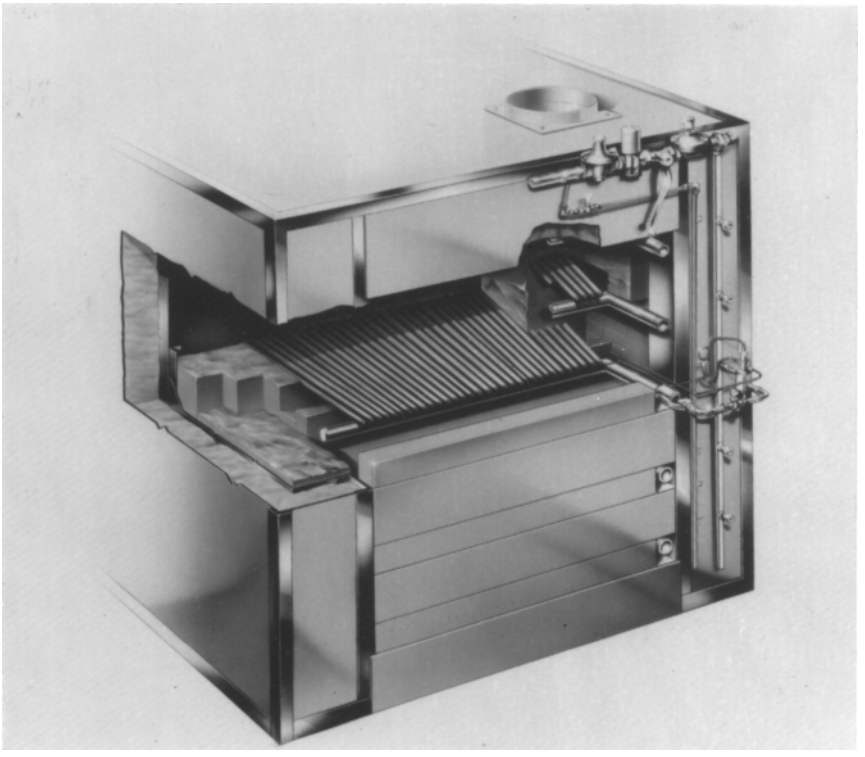


Fig. 16.1 Indirectly heated batch oven.
(Courtesy of Thomas Collins Ltd.)

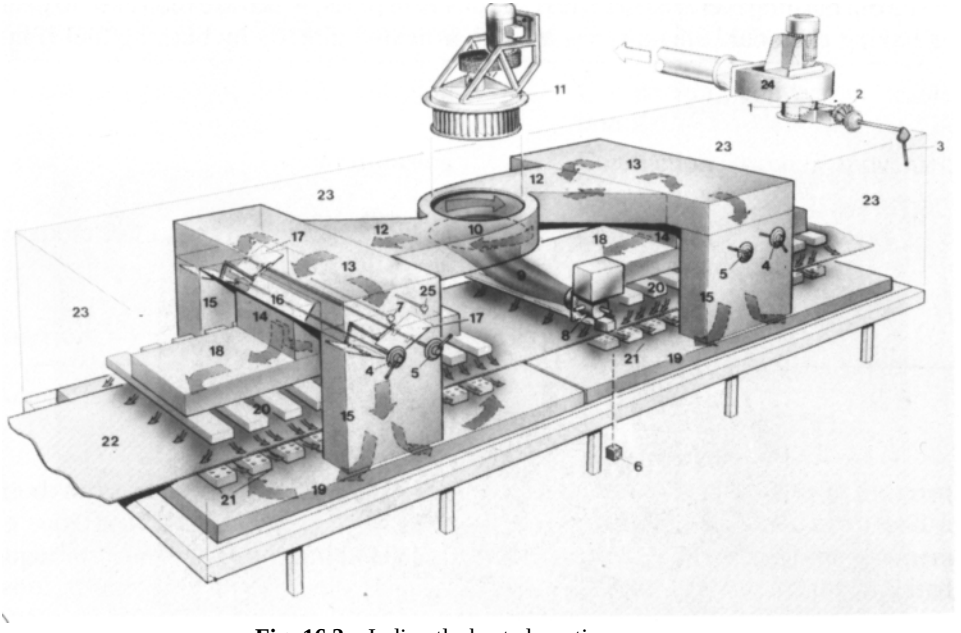


Fig. 16.2 Indirectly heated continuous oven.
(Courtesy of Spooner Industries Ltd.)

belt. Most ovens have 25 mm thick ceramic tiles fitted to the hearth to promote even heat distribution.

Forced-convection hot-air systems have shorter start-up times and a faster response to temperature control than do radiant ovens, because only the air is heated. Conventional heating, forced-convection heating, infrared heating and combined heating methods are compared by Malkki *et al.* (1984). Steam-heated batch ovens are also used for cooking meat products. Similar designs are fitted with smoke generators for smoking meats, cheeses and fish. These are described in detail by Toth and Potthast (1984). The techniques used in smoking are discussed by Lee (1983).

16.2.3 Batch ovens

In the *Peel oven*, food is loaded into a baking chamber, either on trays or singly, by means of a long-handled shovel (a *peel*) which gives its name to the oven. More recent designs include the *multi-deck oven* (Fig. 16.3) which is widely used for baked goods, meats and confectionery products. Some designs have a 'modular' construction to allow expansion of production by duplication of modules, without having to replace the entire plant. The main disadvantages of batch ovens are higher labour costs and lack of uniformity in baking times, caused by the delay in loading and unloading.

16.2.4 Continuous and semi-continuous ovens

Rotary-hearth ovens (Fig. 16.4(a)), *reel ovens* (Fig. 16.4(b)) and *multi-cycle tray ovens* (Fig. 16.4(c)) all circulate the food through the oven on trays, and loading and unloading take place though the same door. The operation is semi-continuous when the oven must be stopped to remove the food. The movement of food through the oven, with or without



Fig. 16.3 Multi-deck oven.
(Courtesy of Werner and Pfeiderer Ltd.)

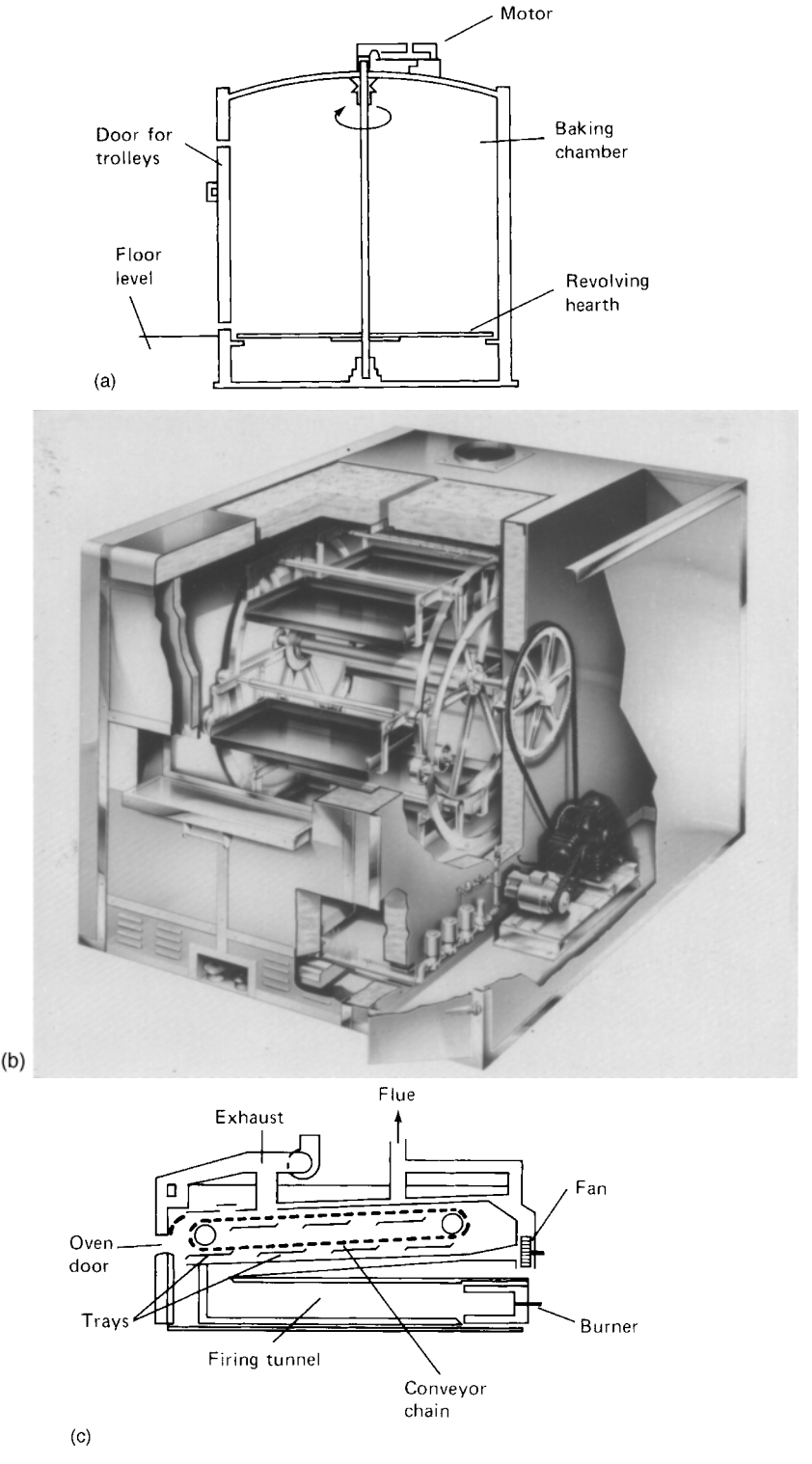


Fig. 16.4 Ovens: (a) revolving hearth oven; (b) reel oven (courtesy of Thomas Collins Ltd); (c) multi-cycle tray oven.
(After Matz (1972).)

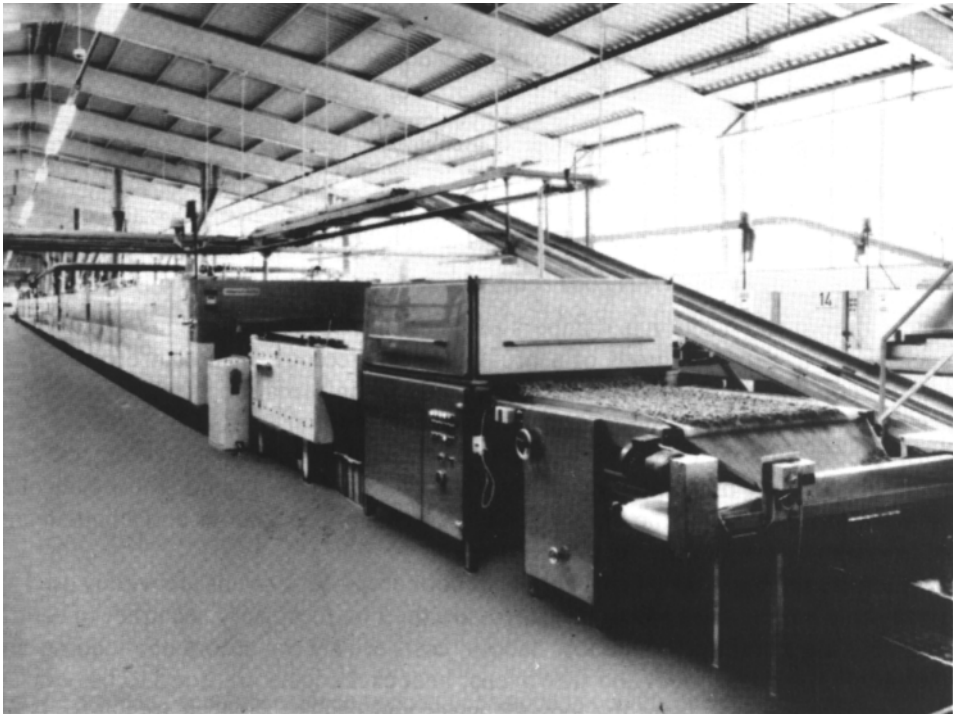


Fig. 16.5 Tunnel oven.
(Courtesy of Werner and Pfeiderer Ltd.)

fans to circulate the air, ensures more uniform heating. *Rotary-hearth ovens* have short baking times but take up a large floor space. *Reel ovens* move the product vertically through the oven and also horizontally from front to back. This permits a larger baking area for a given floor space and more uniform temperature distribution through the oven. The disadvantages of these ovens include the absence of zones of heating, and difficulty in automating loading and unloading. In many applications they are now replaced by tray and tunnel ovens. *Tray ovens* have a similar design to tunnel ovens but have metal trays permanently fixed to a chain conveyor. Each tray holds several baking pans and is pulled through the oven in one direction, then lowered onto a second rack, returned through the oven and unloaded (Fig. 16.4(c)).

Tunnel ovens consist of a metal tunnel (up to 120 m long and 1.5 m wide) (Fig. 16.5) through which food is conveyed either on steel plates (in a *travelling-hearth* type oven) or on a solid, perforated or woven metal belt in the *band* type oven. The oven is divided into heating zones and the temperature and humidity are controlled independently in each zone by heaters and dampers. These retain or remove moisture by adjusting the proportions of fresh and recirculated air in the oven. Vapour (and in direct heating ovens, the products of combustion) are extracted separately from each zone. Many designs are equipped with heat recovery systems (Fig. 16.6).

Microprocessor control of the belt speed, heater output and position of dampers automatically adjusts the baking conditions in each zone, to produce foods of a predetermined colour or moisture content. Details of automatic colour monitoring of baked products are given in Chapter 3 (Section 3.2.2). Microprocessors also provide management information of production rates, energy efficiency and maintenance requirements (Chapter 2). Some ovens are fitted with programmable cycles in which temperature and

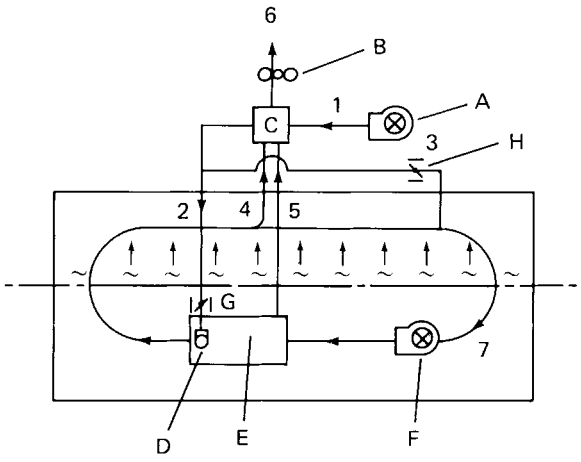


Fig. 16.6 Heat recovery for convection oven: (A) supply air fan; (B) exhaust fan; (C) heat recovery heat exchanger; (D) burner; (E) oven chamber heat exchanger (not in direct-fired ovens); (F) oven chamber air recirculating fan; (G) combustion air control damper; (H) zone integrity air control damper; (1) cold supply air; (2) hot combustion air; (3) hot-zone integrity air; (4) oven exhaust air plus product evaporation; (5) hot-oven heat exchanger exhaust; (6) cooled combined exhaust; (7) recirculating oven chamber air.
(Courtesy of Baker Perkins Ltd.)

time of heating, relative humidity, cooling time and air speed are programmed independently for each of 20 or more products. This allows rapid changes to baking conditions and a high degree of flexibility for different types of product.

Despite the high capital cost and large floor area, these ovens are widely used for large-scale baking. The main advantages are their high capacity, accuracy of control over baking conditions and low labour costs owing to automatic loading and unloading. In both tunnel and tray ovens, heat exchangers are fitted to the exhaust flues to remove heat from the exhaust gases and to heat fresh or recirculated air. Energy savings of 30% are achieved and start-up times can be reduced by 60%.

16.3 Effect on foods

The purpose of baking and roasting is to alter the sensory properties of foods, to improve palatability and to extend the range of tastes, aromas and textures in foods produced from similar raw materials. Baking also destroys enzymes and micro-organisms and lowers the water activity of the food to some extent (Chapter 1, Table 1.4), thereby preserving the food.

16.3.1 Texture

Changes in texture are determined by the nature of the food (moisture content and the composition of fats, proteins and structural carbohydrates (cellulose, starches and pectins)) and by the temperature and duration of heating. A characteristic of many baked foods is the formation of a dry crust which contains the moist bulk of the food (for example meats, bread, potato or yam). Other foods (for example biscuits) are baked to a lower moisture content, and in these the changes that take place in the crust occur throughout the food.

When meat is heated, fats melt and become dispersed as oil through the food or drain out as a component of 'drip losses'. Collagen is solubilised below the surface, to form gelatine. Oils are dispersed through the channels produced in the meat. Proteins become denatured, lose their water-holding capacity and contract. This forces out additional fats and water and toughens and shrinks the food. Further increases in temperature cause destruction of micro-organisms and inactivation of enzymes. The surface dries, and the texture becomes crisper and harder as a porous crust is formed by coagulation, degradation and partial pyrolysis of proteins. In cereal foods, changes to the granular structure of starch, gelatinisation and dehydration produce the characteristic texture of the crust.

Rapid heating produces an impermeable crust which seals in moisture and fat and protects nutrients and flavour components from degradation. A steep moisture vapour gradient is formed between the moist interior (high a_w) and hygroscopic exterior (low a_w) of the food. During storage, moisture migrates through the food. If preservative methods (for example freezing) are not used to immobilise the moisture, this migration softens the crust, lowers the eating quality and thus limits the shelf life of the food. Slower heating permits larger quantities of moisture to escape from the surface of the food before it is sealed by the crust. This results in a shallower moisture vapour gradient and a drier interior in the food. In bread ovens, initial steam heating reduces dehydration of the dough surface, and the crust therefore remains elastic for longer. Ragged breaks in the dough are avoided and the dough expands more fully. The crust is smoother and glossier, and starch hydrolysis promotes Maillard browning reactions to give a better crust colour.

16.3.2 Flavour, aroma and colour

The aromas produced by baking are an important sensory characteristic of baked goods. The severe heating conditions in the surface layers of food cause Maillard browning reactions between sugars and amino acids. Details of the chemistry of the Maillard reaction and Strecker degradation are discussed by a number of workers including Mauron (1982), Danehy (1986) and Fennema (1996). The high temperatures and low moisture contents in the surface layers also cause caramelisation of sugars and oxidation of fatty acids to aldehydes, lactones, ketones, alcohols and esters. The Maillard reaction and Strecker degradation produce different aromas according to the combination of free amino acids and sugars present in a particular food. Each amino acid produces a characteristic aroma when heated with a given sugar, owing to the production of a specific aldehyde. Different aromas are produced, depending on the type of sugar and the heating conditions used (for example the amino acid proline can produce aromas of potato, mushroom or burnt egg, when heated with different sugars and at different temperatures). An abbreviated summary of the aromas produced from the most important amino acids in four foods is shown in Table 16.2.

Further heating degrades some of the volatiles produced by the above mechanisms to produce burnt or smoky aromas. There are therefore a very large number of component aromas produced during baking. The type of aroma depends on the particular combination of fats, amino acids and sugars present in the surface layers of food, the temperature and moisture content of the food throughout the heating period and the time of heating. Details of the aromatic chemicals produced during the baking of bread and roasting of peanuts, coffee, cocoa and meat are given by Adrian (1982).

The characteristic golden brown colour associated with baked foods is due to Maillard reactions, caramelisation of sugars and dextrins (either present in the food or produced by hydrolysis of starches) to furfural and hydroxymethyl furfural, carbonisation of sugars, fats and proteins.

Table 16.2 Aromas produced by baking or roasting

Food	Predominant amino acids	Selected characteristic aromas after heating with a single sugar
Potato	Asparagine	–
	Glutamine	Caramel, butterscotch, burnt sugar
	Valine	Fruity, sweet, yeasty
	Aminobutyric acid	Caramel, maple syrup, nutty
Peanut	Alanine	Caramel, nutty, malt
	Phenylalanine	Sweet and rancid caramel, violets
	Asparagine	–
	Arginine	Bready, buttery, burnt sugar
Beef	Valine	Fruity, sweet, yeasty
	Glycine	Caramel, smoky, burnt
	Leucine	Toasted, cheesy, malt, bready
Cocoa bean	Leucine	Toasted, cheesy, malt, bready
	Alanine	Caramel, nutty, malt
	Phenylalanine	Sweet and rancid caramel, violets
Valine		Fruity, sweet, yeasty

Adapted from Adrian (1982).

16.3.3 Nutritional value

Some baked foods (for example bread and meat) are important components of the diet in many countries and are therefore an important source of proteins, vitamins and minerals. For example, lysine is the limiting amino acid in wheat flour and its destruction by baking is therefore nutritionally important. Other baked foods (for example nuts, biscuits, cocoa, coffee and snackfoods) are less important in the diet, and nutritional losses are therefore less significant.

The main nutritional changes during baking occur at the surface of foods, and the ratio of surface area to volume is therefore an important factor in determining the effect on overall nutritional loss. In pan bread, only the upper surface is affected and the pan

Table 16.3 Vitamin losses in roast meats

Vitamin	Vitamin loss (%)			
	Oven temperature, 150ºC		Oven temperature, 205ºC	
	Beef; internal temperature, 80ºC	Pork; internal temperature, 84ºC	Beef; internal temperature, 98ºC	Pork; internal temperature, 98ºC
Thiamin				
Meat only	39	36	53	46
Drip loss	94	83	–	–
Pantothenic acid				
Meat only	27	35	40	37
Drip loss	80	75	–	–
Niacin				
Meat only	24	31	29	33
Drip loss	84	74	–	–
Riboflavin				
Meat only	25	27	32	31
Drip loss	84	81	–	–

Adapted from Cover *et al*. (1949).

protects the bulk of the bread from substantial nutritional changes. With the exception of vitamin C, which is added to bread dough as an improver and is destroyed during baking, other vitamin losses are relatively small. In chemically leavened doughs the alkaline conditions cause the release of niacin which is bound to polysaccharides and polypeptides and therefore increase its concentration (Appendix B). The vitamin content of bread is also determined by the extent of fermentation which increases the amount of B vitamins (Chapter 7). In meats, nutrient losses are affected by the size of the piece, the type of joint, the proportions of bone and fat, pre- and post-slaughter treatments and the type of animal. Some thiamine is removed in pan drippings but, as these are usually consumed, the overall losses are smaller. Cover *et al.* (1949) studied the effect of baking temperature on vitamin losses in different meats. At 150ºC the meats were well cooked and total thiamine losses were moderate. At higher temperatures the pan drippings were charred and inedible, and total losses were therefore substantially increased (Table 16.3).

In biscuits, breakfast cereals and crispbread the bulk of the food is heated to a similar extent. However, these are smaller pieces which require a shorter baking time, and losses are therefore reduced. In prepared foods, which have ingredients that have been processed to stabilise them for storage, there may be additional losses in nutritional quality (for example from milling wheat, drying fruit, frozen storage of meats or fermentation and drying of cocoa and coffee beans).

Thiamine is the most important heat-labile vitamin in both cereal foods and meats, and losses are reported in Table 16.4. In cereal foods the extent of thiamine loss is determined by the temperature of baking and the pH of the food. Loss of thiamine in pan bread is approximately 15% (Bender, 1978) but in cakes or biscuits that are chemically leavened by sodium bicarbonate, the losses increase to 50–95%.

During baking, the physical state of proteins and fats is altered, and starch is gelatinised and hydrolysed to dextrins and then reducing sugars. However, in each case the nutritional value is not substantially affected. The loss of amino acids and reducing sugars in Maillard browning reactions causes a small reduction in nutritive value. In particular, lysine is lost in Maillard reactions, which slightly reduces the protein quality. In bread the *protein efficiency ratio* is reduced by 23% compared with that of the original flour (Bender, 1978). The extent of loss is increased by higher temperatures, longer baking times and larger amounts of reducing sugars. The amylase activity of flour, the addition of sugar to dough, the use of fungal amylases (Chapter 7), and steam injection into ovens to gelatinise the surface starch and to improve crust colour all therefore affect the nutritive value of the proteins to some extent. In biscuits, a reduction in dough thickness from 4.9 mm to 3.8 mm, each baked at 170ºC for 8 min, produced higher losses

Table 16.4 Thiamin losses during baking

Food	Thiamin loss (%)
Beef	40–60
Pork	30–40
Ham	50
Lamb	40–50
Poultry	30–45
Bread	15
Cake	23
Cake[a]	30–95
Soya bean	90

[a] Chemical leavening agent used.
Adapted from Farrer (1955).

of amino acids as follows: tryptophan, from 8% to 44%; methionine, from 15% to 48%; lysine, from 27% to 61% (Mauron *et al.*, 1960). In maize, lysine loss is increased from 5% to 88% during the manufacture of breakfast cereals, which is corrected by fortification.

16.4 Acknowledgements

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